

Protactinium

91 Pa The name protactinium comes from the Greek word *protos*—meaning “first”—coupled with actinium. It refers to the fact that protactinium is formed before actinium, in a radioactive decay sequence that starts with uranium. A form of protactinium is used by scientists to calculate the age of ocean sands.



Lise Meitner

Protactinium was initially named brevium from the Latin word for “brief” because it was unstable. In 1918, Austrian physicist Lise Meitner (left), working with Otto Hahn, found a more stable form and named the new element protactinium.

Torbernite

This mineral contains copper and uranium. The uranium present decays, eventually releasing protactinium in small quantities. The vivid green crystals are popular with mineral collectors, but they can release radioactive radon gas, which is a major health hazard.



Neptunium

93 Np The discovery of neptunium in 1940 was an important one, as it was the first element after uranium to be artificially created. Its discovery opened up a new world of transuranium, or transuranic, elements. For their work in this area, the 1951 Nobel Prize in Chemistry was awarded to the US scientists Edwin McMillan and Glenn T. Seaborg.

Uraninite

A naturally occurring mineral, uraninite was originally known as pitchblende. It is composed chiefly of uranium compounds but also contains small amounts of some other elements, including neptunium.



Americium

95 Am Another artificially produced radioactive actinide, americium was first synthesized in the US in 1944. This element is only produced in nuclear reactors. It also emits rays that can be used to monitor the thickness of metal sheets in industry.

Detecting smoke

Americium's main commercial use is in detectors in smoke alarms. The safe radiation given off by americium allows an electrical current to flow, but if smoke enters the detector, its particles absorb the radiation, the circuit is broken, and an alarm sounds. Once the smoke clears, the current flows again, and the alarm turns off.

